Response to Reviewers — BIOADV-2026-296 (revised resubmission)

We thank the reviewers and the Associate Editor for a rigorous and constructive review. The decision turned on one question — *is the ranking result general, or is it an artifact of this particular comparison?* — and we have tried to answer it with experiments rather than argument.

What is new, and why it answers the decision

Four additions carry the revision. We put them first because each targets the generality concern directly.

1. **A nine-model roster (§4.4, Table 3).** The sharpest form of the objection is that a claim about "rankings" made over four models is a claim about nothing. We widened the comparison to nine learners spanning memorization propensity end to end — 1- and 15-nearest neighbours, random forest, extremely randomized trees, a multilayer perceptron, gradient boosting, linear SVM, logistic regression, Gaussian naive Bayes — all trained from scratch on the same splits with the same decision rule. The result is stronger than the four-model version and sharper: **the leaky split cannot separate its top three models at all** (0.8736 / 0.8729 / 0.8721, within 0.0015, paired interval on the top-two margin including zero) **yet those same three span 23 accuracy points once corrected** (0.768 / 0.537 / 0.628). 1-NN falls from rank 2 to rank 9, 0.873 → 0.537. The benchmark is not merely crowning the wrong model; it is blind to enormous real differences between models it reports as tied. The corrected winner is the linear SVM — the same model the four-model comparison selects, so the two analyses agree. Four predictions (P1–P4) were committed before the run and all four are scored in the text whether or not they held.
2. **A construct-and-break manipulation (§4.7, Table 4).** The previous submission argued the cause was a curation defect; it did not *test* it. We now intervene on the construction step alone, in both directions. Applying the omitted merge step to `human_enhancers_ensembl` collapses its leak fraction 0.390 → 0.012, its random-forest inflation +0.165 → −0.008, and the reordering disappears. Conversely, imposing the same defect on the clean `human_ocr_ensembl` — at matched size *and* balance — raises leakage 0.004 → 0.204, inflation +0.002 → +0.105, and **manufactures the reordering on demand** (RF promoted 1st on the contaminated split, demoted to 4th by correction). Model code, split ratio and (in the fix-a-leaky direction) the sequences themselves are untouched; the intervention is on the construction step alone. This converts the causal claim from correlational to interventional.
3. **A cross-suite census (§4.10) — and an honest null.** We applied the census unchanged to the Nucleotide Transformer downstream tasks. Three of eleven independent tasks are leaky — one at 25.0% *byte-identical* train/test overlap, verified on a separate code path using no k-mers — and six more borderline. **No ranking inverts materially on any of them, and we report that null prominently rather than burying it.** One swap does occur — HistGradientBoosting overtakes the forest on `enhancers` after correction, 0.887 vs 0.869 — but the two were separated by only 0.005 on the as-shipped split, so it is immaterial under our own inversion criterion, and we say so rather than counting it either way. The null is not an embarrassment but a test passed: our diagnostic condition (§3.13) predicts it in advance from the as-shipped split, because reordering requires not just leakage but a challenger that is genuinely better on novel sequences, and on 22 of 24 pairs the as-shipped leader is also the novel-sequence leader. On the two pairs where the condition had something to say it was right both times

— including the one case a trivial "nothing swaps" baseline gets wrong. Only one of those two is admissible under our own Methods §3.13, which permits the condition only where the intermediate similarity band is small: the `splice_sites_acceptors` pair's band is 33.3%, so we rule that call out and do not count it. The condition therefore rests on **one** admissible out-of-sample call, and we state it that way in §4.10 rather than claiming the two. We also state plainly that the aggregate 24/24 score is nearly uninformative, since that baseline scores 23/24.

4. **Honest rescoping of the central claim.** The previous "changes which model wins on 2/2 leaky datasets" claim was too strong. On `human_nontata_promoters` the demotion holds only under accuracy@0.5: it reverses under AUROC and F1, is a CI-overlap tie, does not reproduce under chromosome holdout, is milder under alignment, and the prevalence-corrected graded gap does not separate the model families there at all. The ranking-change result is now claimed **cleanly for `human_enhancers_ensembl`** (an existence proof, holding on every axis tested) and **as an explicit cautionary partial case for `human_nontata_promoters`**. The title, abstract, Discussion and new Conclusion all carry that scope.

Three further structural changes: the paper adopts **near-duplicate leakage** as its primary term and reserves *homology* for where alignment or containment evidence earns it (R3.4); it adds a **Conclusion** (§6); and it now reports its **software environment** with pinned versions (§3.14).

Associate Editor (Prof. Shanfeng Zhu)

The Associate Editor summarized five overarching concerns; each is addressed and cross-referenced below.

- **Limited model scope.** Answered twice over: the nine-model roster above (§4.4), and a from-scratch 1D residual CNN with a pre-registered dropout×weight-decay grid and a binding refutation condition (§4.9). The named pretrained foundation models are scoped out with a *quantified* ≈100% pretraining-overlap confound rather than an asserted one (R1.1, R2.a3, R3.1).
- **Validation of homology detection.** MMseqs2 alignment identity reproduces the effect metric-independently, and a length-robust containment index is added — which honestly flags one previously-clean dataset as borderline (R1.3, R2.a2, R3.4).
- **Bootstrap methodology.** A cluster (block) bootstrap over within-test near-duplicate components, with intraclass correlation, design effect, an analytic cluster-robust cross-check and a combined-source interval; it changes no significance verdict (R3.3).
- **Dataset characteristics.** Class balance is disclosed as curated, not natural, and a prevalence-aware imbalanced evaluation is added (R2.a1).
- **Hyperparameter choice.** A random-forest regularization path shows the effect is a property of the *default unregularized* forest — and, critically, that cross-validation on the as-shipped training set *selects* that configuration (R3.2, and see the Discussion).

Reviewer 1

R1.1 — "The evaluation uses only classical models; a trained deep network is needed, and the reach beyond this domain is unclear."

(a) Deep model. We added a **from-scratch 1D residual CNN** in the Basset/DeepSEA lineage (one-hot input, no k-mer features, trained on each training split only — no pretraining confound), with a **pre-registered** dropout×weight-decay grid and a binding refutation condition pre-specified in a timestamped document in the code release (`results/deep_preregistration.md`). The standard-practice regularized reference cell (dropout 0.3, weight decay 1e-4) drops on **both** leaky datasets — `human_nontata_promoters` **+0.043** (cluster-bootstrap CI [0.025, 0.061]), `human_enhancers_ensembl` **+0.014** ([0.007, 0.021]), both excluding zero — so the pre-registered refutation condition is **not** triggered. Nineteen of the 20 cells drop with cluster CIs excluding zero; the sole exception is one nontata cell (dropout 0.6 / wd 1e-4) where heavy dropout underfits, leaving no inflation to lose. Per-cell single-run variance is non-trivial, so we read the mechanism from the grid-level contrast, and the unregularized manipulation-check cell memorizes hardest (ensembl graded gap **+0.222**, mirroring the unregularized RF). The effect is therefore **not** a classical-model artifact. New **§4.9**; Methods **§3.8**.

We also report one pre-registered expectation that was **not** met: the pre-registration called this a *dose-response* grid, and it is not one — the drop is non-monotone in both dropout and weight decay. We state this rather than quietly restating the grid, since a pre-registration reported selectively is worth nothing.

(b) Reach. We now separate two claims explicitly (Introduction): **Claim I** (de-leaking lowers accuracy) is credited to prior work across domains and reproduced only as a within-suite control; **Claim II** (leakage can *reorder* rankings) is our contribution, scoped to short human regulatory DNA plus a memorization-prone model in the comparison set. The title carries that scope.

R1.2 — "The corrected ranking is treated as the true generalization ranking, but de-duplicating the training set also removes learnable signal."

Agreed, and we now **measure** rather than assert it. On `human_nontata_promoters` the corrected and novel-only rankings diverge ($\tau = -0.67$): the random forest is demoted under the corrected split yet remains best on truly novel sequences, so its advantage is partly genuine generalization. Single-linkage **cohesion** supplies the biological reason (§4.13): ensembl's clusters are 99.6% pairwise with largest-cluster cohesion 0.96 — clean discrete locus duplication, where correction is a clean fix — whereas nontata's 420-member largest cluster has cohesion 0.083, i.e. heavy transitive chaining that over-removes genuine promoter and gene-family signal. We reframe both the corrected split and novel-only accuracy as explicit near-duplicate-bounded proxies (Methods §3.5, §3.9) and name leave-one-chromosome-out as the deployment-relevant control (§4.12). This is the core of the nontata-as-cautionary-case framing (§4.3, §5).

We also disclose a decomposition that cuts the other way and belongs in the record (§4.3): on ensembl, evaluated on the as-shipped split's *novel* stratum with no re-splitting and no retraining, the forest already ranks below the linear SVM, but by only 1.2 points; the remaining $\sim$10 points are a **retraining penalty**, because de-leaking removes near-duplicate partners from the training set too. Both are consequences of leakage, but they are different quantities and we now say so.

R1.3 — "Validate leakage with an alignment-based measure and give threshold recommendations."

Alignment: we re-clustered both leaky datasets with **MMseqs2** alignment identity, fed the external clusters into the *same* whole-cluster splitter (only the edge metric changes), and refit all four models. The `human_enhancers_ensembl` effect is metric-independent (RF drop **0.164** Jaccard ≈ **0.162** alignment; RF still rank 4); nontata is milder under alignment (0.108 → 0.064), reinforcing its partial status. We describe this as "alignment-scored", not k-mer-independent, since MMseqs2 is itself k-mer-seeded (**§4.12**, Methods **§3.6**).

Thresholds: the clustering-threshold sweep (0.5/0.7/0.9) changes no verdict (Methods §3.5); the length-robust containment re-measure and the bimodal-vs-graded similarity structure (cohesion 0.96 vs 0.083) provide the decision basis. The new Conclusion (§6) states the resulting recommendations plainly: audit at full dataset scale, report both a length-blind and a length-robust similarity, deduplicate in coordinate space before splitting, and keep a memorization-prone learner in the comparison set as a tripwire.

Reviewer 2

R2.a1 — "Class balance is not characterized, and imbalanced data are not tested."

We now disclose that every balanced dataset is **curated** to a 0.500 positive fraction in both train and test (positive/negative subfolders); only nontata is a natural mild imbalance (0.544) (Methods §3.1). Because balance is preprocessed, we add a **prevalence stress test** (RF, $\pi \in \{0.5, 0.2, 0.1\}$) with prevalence-aware metrics: the leakage inflation is **revealed by AUPRC/MCC/minority-recall but masked by accuracy** (ensembl π=0.2: AUPRC 0.622 → 0.477, MCC 0.422 → 0.182, minority recall 0.258 → 0.070, while accuracy barely moves, 0.844 → 0.808). The splitter is extended with a per-class realized-fraction check and preserves the target prevalence (realized 0.5001 / 0.2007 / 0.1001). **§4.13**; Methods §3.9.

R2.a2 — "The similarity measure and the features are both k-mer based, so the leakage finding may be circular."

Addressed by the alignment re-measure (R1.3): with alignment identity as the edge metric and the same splitter, the ensembl RF drop is essentially unchanged (0.164 vs 0.162) and the demotion persists. We add two further metric-independent lines of evidence. First, ensembl's leakage is **38.0% byte-identical** duplication — no similarity metric is involved in counting exact copies. Second, and new in this revision, the **coordinate-space analysis** (§4.6) identifies the defect in the shipped genomic intervals, which are causally upstream of and statistically independent from the k-mer clustering: 77,421 positive intervals sit on only 40,934 distinct coordinates, and every one of the 36,487 duplicated pairs carries byte-identical sequences and a single label. That is a cross-modal confirmation, not a restatement. **§4.12**, **§4.6**; Methods §3.4, §3.6, §3.10.

R2.a3 — "Extend to deep models and to the multiclass setting."

Deep models: the CNN (R1.1a, §4.9) and the multilayer perceptron in the nine-model roster (§4.4), whose drop also excludes zero — the inflation reaches a neural learner that we had not designated memorization-prone in advance. **Multiclass:** the only multiclass set is clean, so we answer by construction, injecting label-carrying near-duplicates at f ∈ {0, 0.1, 0.2, 0.4} into the clean 3-class `human_ensembl_regulatory` set. The RF drop grows monotonically with f (+0.0004, +0.118, +0.179, +0.258) while LR stays flat (≤0.066), and the corrected accuracy is stable (∼0.58) because the split removes the injected leakage. **§4.13**; Methods §3.9. We state explicitly that leakage is injected because the suite ships no naturally leaky multiclass set.

R2.b1 — "Give a threshold-sensitivity analysis and concrete recommendations."

The verdict is robust to the clustering threshold (0.5/0.7/0.9 sweep, no change; Methods §3.5) and to removal of the single largest component. We report the geometry that drives the recommendation: the length-robust containment index (§4.1, §4.13) as the metric to prefer when lengths vary, and single-linkage cohesion as the diagnostic separating a clean discrete-duplication case (ensembl — one threshold suffices) from a transitive-chaining case (nontata — treat as partial scope, avoid over-removal). Report-card guidance and the length-cap disclosure are in Table 2; the consolidated recommendations are in §6.

Reviewer 3

R3.1 — "Deep models are needed, specifically attention-based / foundation models (e.g. DNABERT-2, HyenaDNA)."

(i) Trained deep net: answered by the from-scratch 1D residual CNN (§4.9), trained on the training split only, whose regularized reference cell drops on both leaky datasets with cluster-bootstrap CIs excluding zero — the memorization mechanism reaches a trained neural network with no pretraining confound. We also attempted a from-scratch attention (Transformer) encoder under the same protocol, and mention it here only so the record is complete: **it is not in the manuscript and we rest no claim on it.** It did not yield a usable comparison — on `human_nontata_promoters` (251 bp) the corrected split slightly *raised* its accuracy (drop −0.016), consistent with a data-hungry architecture underfitting a short, local-motif task, and the `human_enhancers_ensembl` arm did not finish within our compute budget. We therefore report the CNN as the trained-deep-net instrument and leave the from-scratch-attention question open rather than resting an argument on one incomplete run. Separately, the pretraining-contamination argument below justifies excluding *pretrained* attention models, not attention architectures per se.

(ii)/(iii) Named foundation models: DNABERT-2, HyenaDNA and the Nucleotide Transformer are cited and discussed but scoped **out** of the split-effect evidence, and we now **quantify** the confound rather than assert it. The two leaky datasets' sequences are extracted from the human reference genome — we recovered their hg38 chromosomal coordinates to build the chromosome-holdout control — so **≈100% of these test sequences already lie inside HyenaDNA's whole-genome (hg38) pretraining corpus** and within the human component of DNABERT-2's and the Nucleotide Transformer's multispecies corpora. That is a second leakage channel no train/test re-split can close, so fine-tuning them would yield a confounded lower bound rather than split-effect evidence. We therefore leave the deployed-foundation-model question **genuinely open** (§5) rather than answer it with a contaminated number. We note this is not evasion but the same problem one level up: evaluating those models properly requires a pretraining-overlap-aware protocol, which is itself an instance of the leakage problem this paper studies.

R3.2 — "The random forest runs at memorization-maximizing defaults; is the result just an untuned-RF artifact?"

This was the objection we took most seriously, and we answer it empirically.

First, the **regularization path** (min_samples_leaf ∈ {1…500}, max_depth ∈ {None…4}, at full scale, largest settings exceeding the largest near-duplicate cluster). At the default the forest scores 1.000 on the ≥0.9-similarity bin with the full drop (0.164 / 0.108) and graded gap (0.230 / 0.121); regularizing drives the drop to ≈0 (0.164 → −0.001; 0.108 → 0.000), collapses the graded gap, and the "RF wins on the leaky split" phenomenon itself **requires** the unregularized forest — rank 1 flips to rank 4 once min_samples_leaf ≥ 20. **§4.8**; Methods §3.7.

Second, and this is the direct answer: **a practitioner does not know the split is leaky, and tuning does not rescue them.** We cross-validated the forest over min_samples_leaf ∈ {1…500} on the as-shipped *training* set under two schemes — ordinary stratified 5-fold (what a practitioner would actually run) and 5-fold holding whole near-duplicate clusters out together (the honest version). We ran both schemes on **both** leaky datasets.

On `human_enhancers_ensembl`, the dataset carrying the reordering claim, the two procedures diverge and the consequence is stark. Ordinary cross-validation selects **min_samples_leaf = 1**, the memorizing default, and the model it selects scores 0.860 as shipped but 0.696 once the leakage is removed — a loss of 16.4 points. Cluster-grouped cross-validation selects **min_samples_leaf = 20**, and that model loses only 2.2 points (0.751 → 0.729). So tuning does not rescue the practitioner: ordinary tuning actively selects the configuration whose apparent advantage is almost entirely leakage. Worse, the benchmark **punishes the correct methodology** — the leakage-aware tuner's model looks 10.8 points *worse* as shipped while being 3.3 points *better* corrected. On `human_nontata_promoters` both procedures select leaf = 1, so our pre-registered expectation P5 holds on `human_enhancers_ensembl` and ties, rather than reverses, on `human_nontata_promoters`. Discussion §5.

The objection is therefore answered on the dataset where it matters: the untuned default is not a careless choice a tuner would correct, it is *what ordinary tuning selects*.

Finally, we note the objection does not reach the core finding in any case, because §4.7 shows the defect is a property of the **data**: editing coordinates alone, with model code untouched, both removes the reordering and creates it.

R3.3 — "Near-duplicates violate independence; a cluster/block bootstrap is the correct uncertainty estimate."

Implemented and reported in full. Resampling whole within-test near-duplicate components as blocks widens the CIs by **1.0–1.8×** but **flips no verdict**: the RF drop still excludes zero on both datasets (ensembl [0.155, 0.173]; nontata [0.092, 0.126]) and the ensembl LR drop stays null. We now also report the diagnostics the reviewer's request implies: the intraclass correlation is high (0.78–0.99) yet the **design effect is small** (1.06–1.17), because the correlated near-duplicates are test-to-**train**, not test-to-test — within-test clustering is only 1.06–1.19 sequences per component, a regime in which a sample-wise bootstrap of a fixed trained model's test correctness is nearly unbiased. An analytic Liang–Zeger cluster-robust standard error agrees with the block bootstrap, and a **combined-source interval** folding in the five re-split-seed variance still excludes zero for both leaky RF drops (ensembl [0.154, 0.174]; nontata [0.088, 0.129]). Every deep-model drop CI (§4.9) uses this cluster bootstrap. **§4.13**; Methods §3.9.

Relatedly, because we report roughly a dozen leaky-versus-clean delta intervals, we applied a **Benjamini–Hochberg correction** at q = 0.05: it leaves the two leaky RF drops and the nontata LR drop significant and no clean or three-class delta significant, so multiplicity changes no verdict either.

R3.4 — " 'Homology' is the wrong word for near-identical/duplicate leakage."

Agreed and adopted throughout: **near-duplicate leakage** is the primary term, the splitter is **near-duplicate-aware**, and the title is changed. An exact-duplicate census settles the matter metric-independently (`human_enhancers_ensembl` = 38.0% byte-identical copies; `human_nontata_promoters` = 0 exact duplicates yet 22.5% near-duplicates at Jaccard ≥ 0.9 — the more homology-like but statistically weaker case). We keep "homology" only where the alignment or containment re-measure earns it (Methods §3.4), and we downgrade the report card's green verdicts to "no forward-strand near-duplicate leakage detected (length-cap disclosed)" (Table 2, §5), since a clean verdict carries a heavier evidential burden than a leaky one.

Associate Editor — "This cannot be addressed in a short time / new experiments are needed."

We agree it could not be addressed quickly, and did not try to. The revision adds the two hard-gate experiments the panel expected — a trained deep model (§4.9) and alignment-based validation (§4.12) — plus the genomics-standard **leave-one-chromosome-out control** (§4.12), which independently reproduces the marquee `human_enhancers_ensembl` demotion (RF rank 1 → 4, identical corrected order) and independently confirms `human_nontata_promoters` as the partial case (RF stays rank 1). Beyond those, it adds the nine-model roster, the construct-and-break manipulation, the coordinate-space analysis, the cross-suite census with its honest null, the tuning-selection experiment, and the diagnostic condition — six experiments that did not exist in the prior submission.

We also closed three internal issues a careful reviewer would have flagged:

- the central claim is honestly rescoped to `human_enhancers_ensembl` (n=1 clean, partial on the second);
- the previously inconsistent headline drop numbers are reconciled to a single decision rule (`.predict()`/argmax) and estimator (5-seed mean + bootstrap CI), removing the earlier three-coexisting-values spread and the incorrect "thread-nondeterminism" wording — the true causes were a decision-rule inconsistency and a point-vs-mean estimator choice, and we say so;
- every pre-registered prediction is now scored in the text whether or not it held. Of the six, P1 **fails as registered** — the corrected-gap result that favours it was defined after P1 was committed and we label it post hoc rather than count it — P2 holds on one leaky dataset and fails on the other, P3 and P4 hold, P5 holds on `human_enhancers_ensembl` and ties on `human_nontata_promoters`, and P6 holds on all ten doses. We report the failures as prominently as the successes.